

SAHIL AHAMED SHAIK

Robotics enthusiast



- **EDUCATION:**
 - B.tech-mechatronics from Hindustan Institute of Technology and Sciences (HITS).
- **EXPERIENSES:**
 - Abhyaz internship (Robotics intern)
 - Learning about ROS, SolidWorks, and CAD software
 - Self-learning approach.
 - Edgeforce solutions (ROS intern)
 - Worked on UGV path-planning (MOVE_BASE PACKAGE + creating self program)
 - Worked on CBM designing and developing along with UGV parts
 - Worked on EXO skeleton.
- **PUBLICATIONS AND PRESENTATIONS:**
 - Design and evaluation of various knee assistive mechanisms for rehabilitation design and evaluation of various knee assistive mechanisms for rehabilitation, AIP Publishing · Nov 10, 2023
 - Presented at RIACT 2022, RANE 2023 and AIIMS- 3.0.
 - Patent and journal is in process.

PROJECTS DONE

- **Designing and developing of autonomous sanitizing robot**
 - Learned the basic circuit and sensor details
 - The development of robot body also studied
- **Design and development of a homogeneous swarm of cooperative robots in a virtual environment**
 - The robot communication methods explored
 - The inner robot communication is established using the central node
- **Design and development of knee rehabilitation robot**
 - Designed more than five SolidWorks designs and developed three models for knee rehabilitation
 - Infused AI to detect knee osteosis
 - Patent in progress and publication is done
- **Design and Development of Borewell Rescue System**
 - Designed robotic arm in SolidWorks
 - 2 designs have been designed and one design was developed

MAJOR SKILLS

- ROS 1&2 – GAZEBO, RVIZ, MOVEIT
- SolidWorks, Fusion 360
- ANSYS
- Python, C++
- AI(tensor flow)
- Unity (AR/VR)
- MATLAB
- CAD software

CERTIFICATIONS

- Completed 3 days of SMC training in pneumatics.
- Completed TensorFlow 2.0: Deep Learning and Artificial Intelligence-Udemy.
- Completed ROS for Beginners: Basics, Motion, and OpenCV- Udemy.
- Completed Introduction to AI, Machine Learning, and Python basics- Udemy. Completed MATLAB and Simulink onramp- MathWorks training service
- In progress :
 - RPA
 - Reinforcement learning - courses

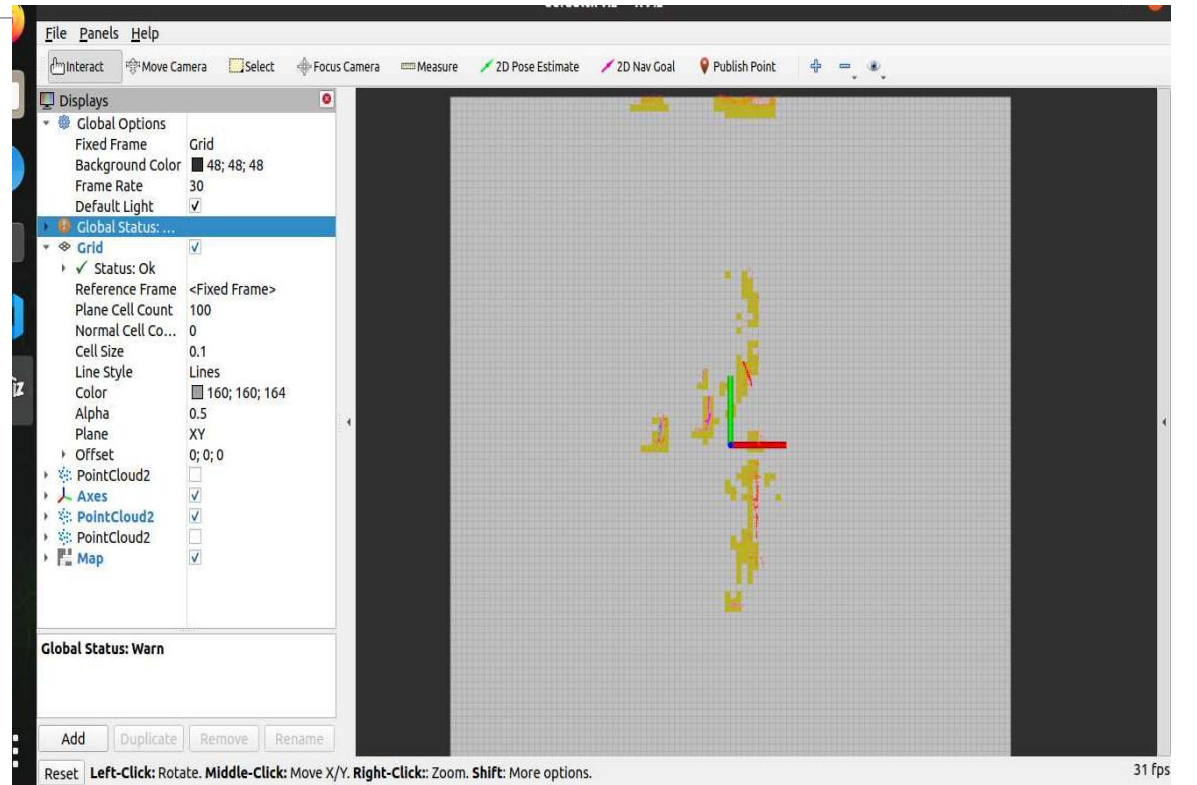
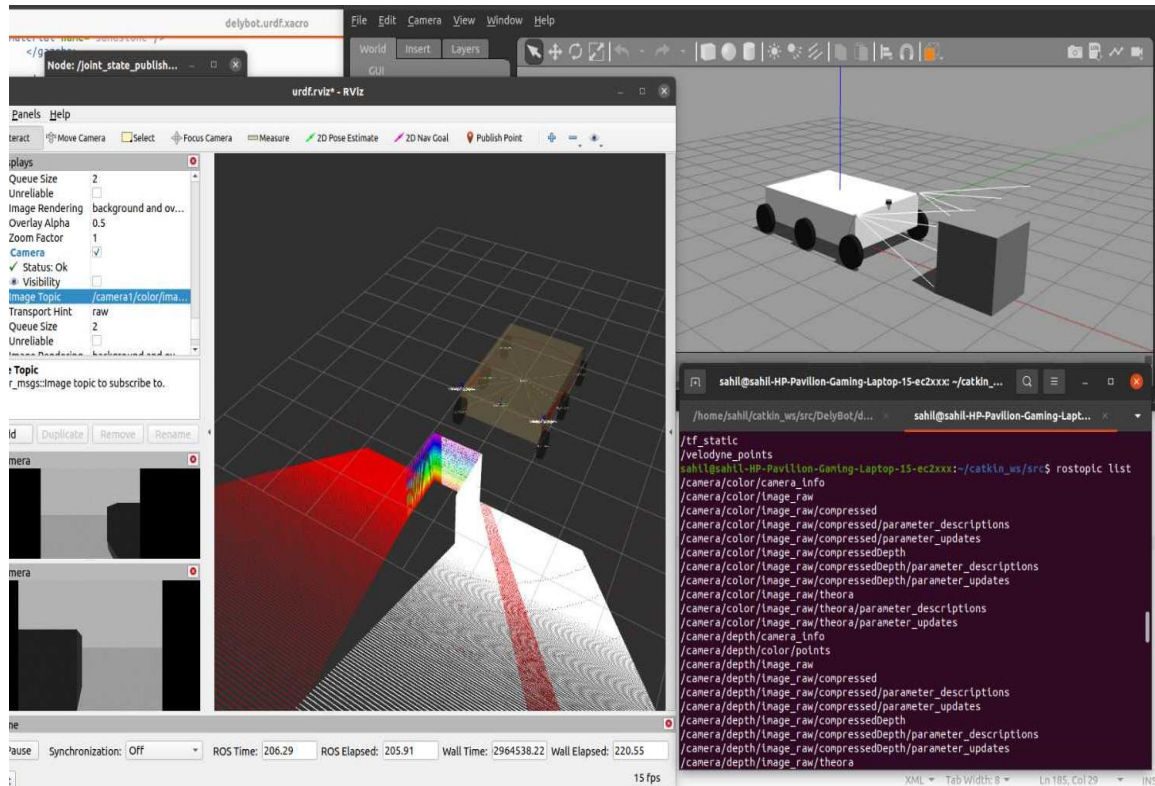
UGV Path-Planning

Edgeforce solutions(experience)

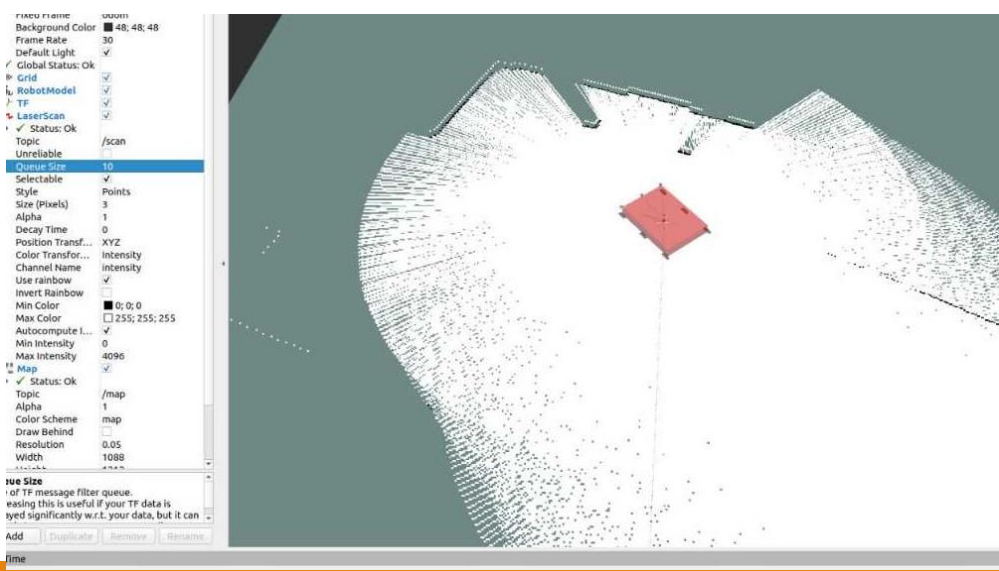
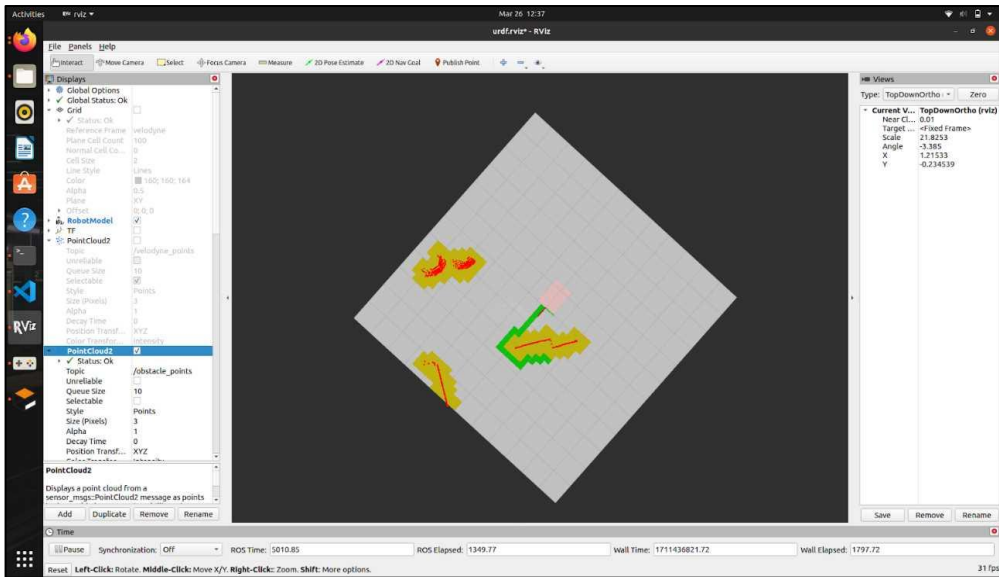
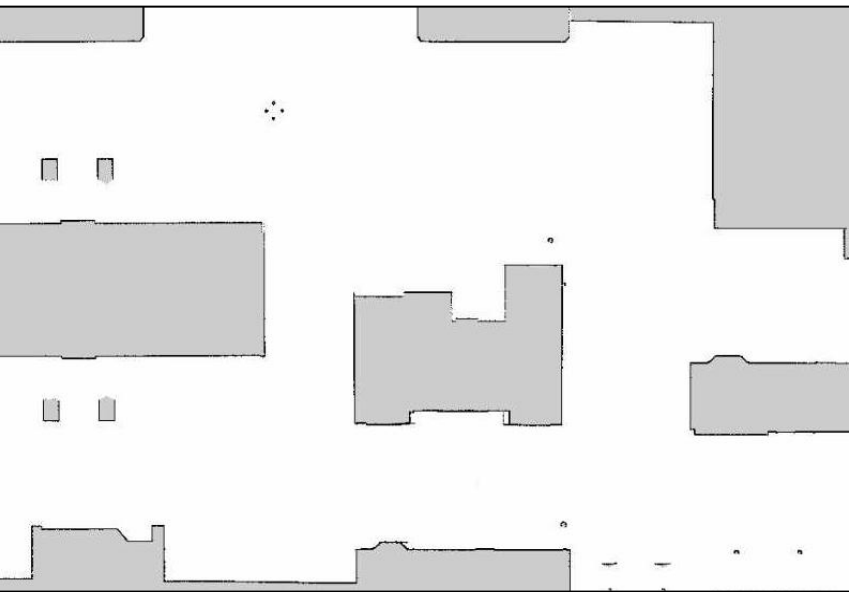
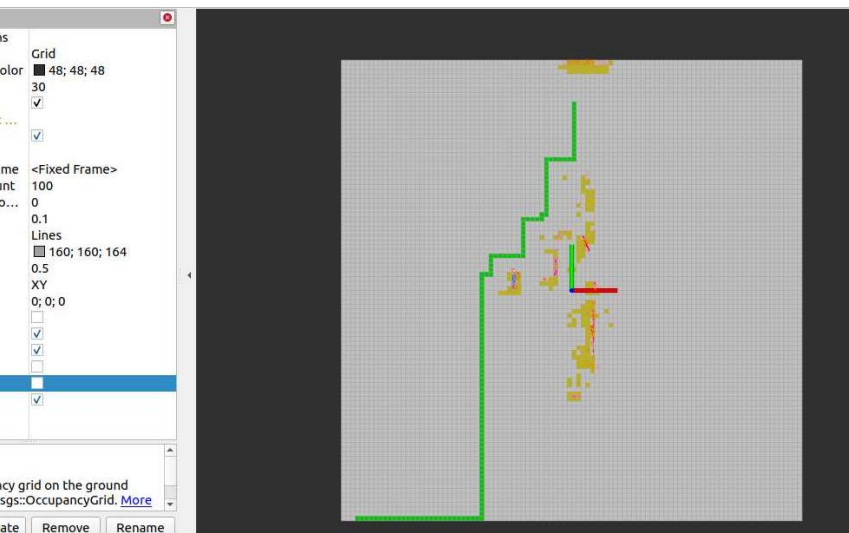
UGV Path-Planning

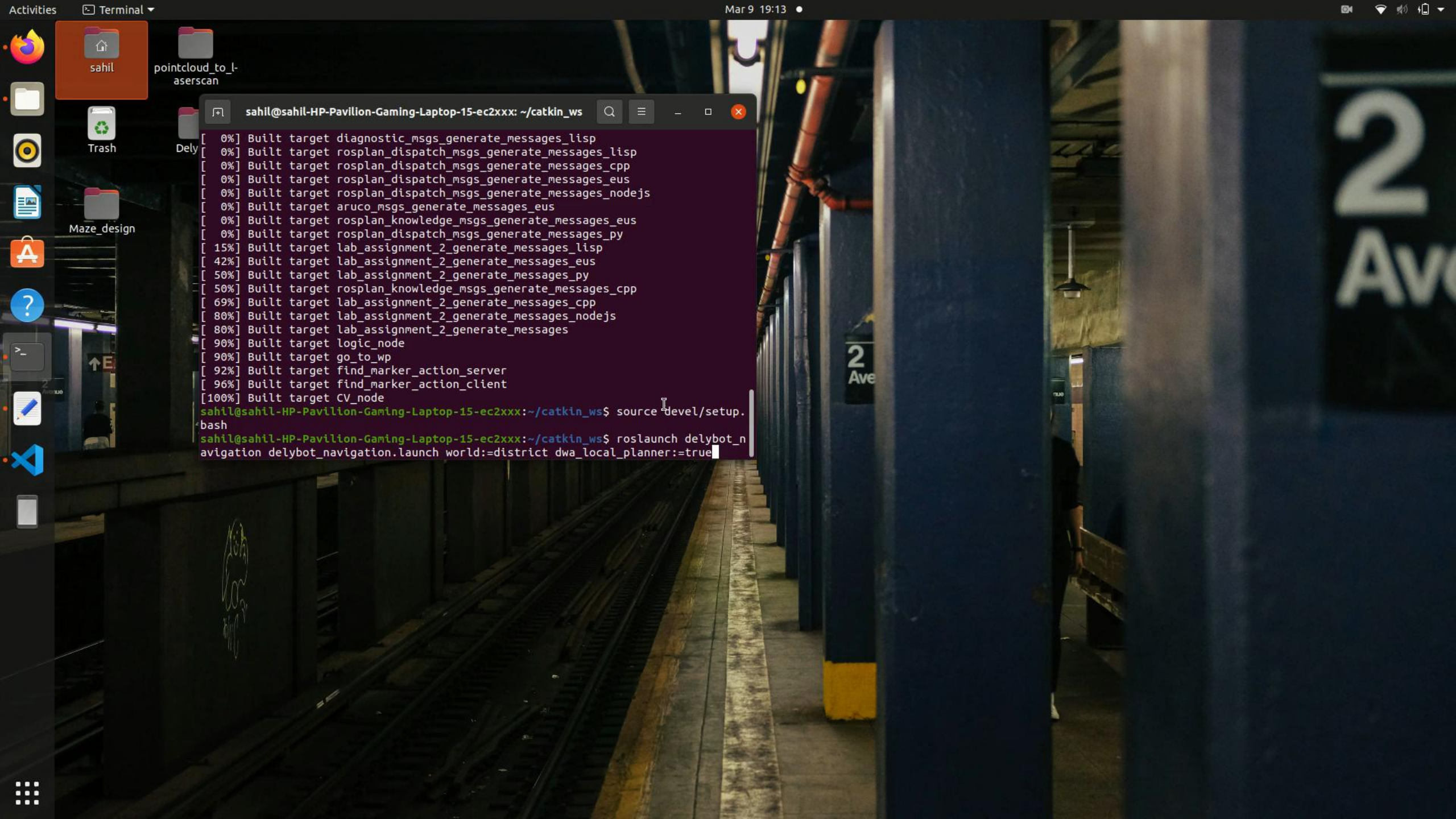
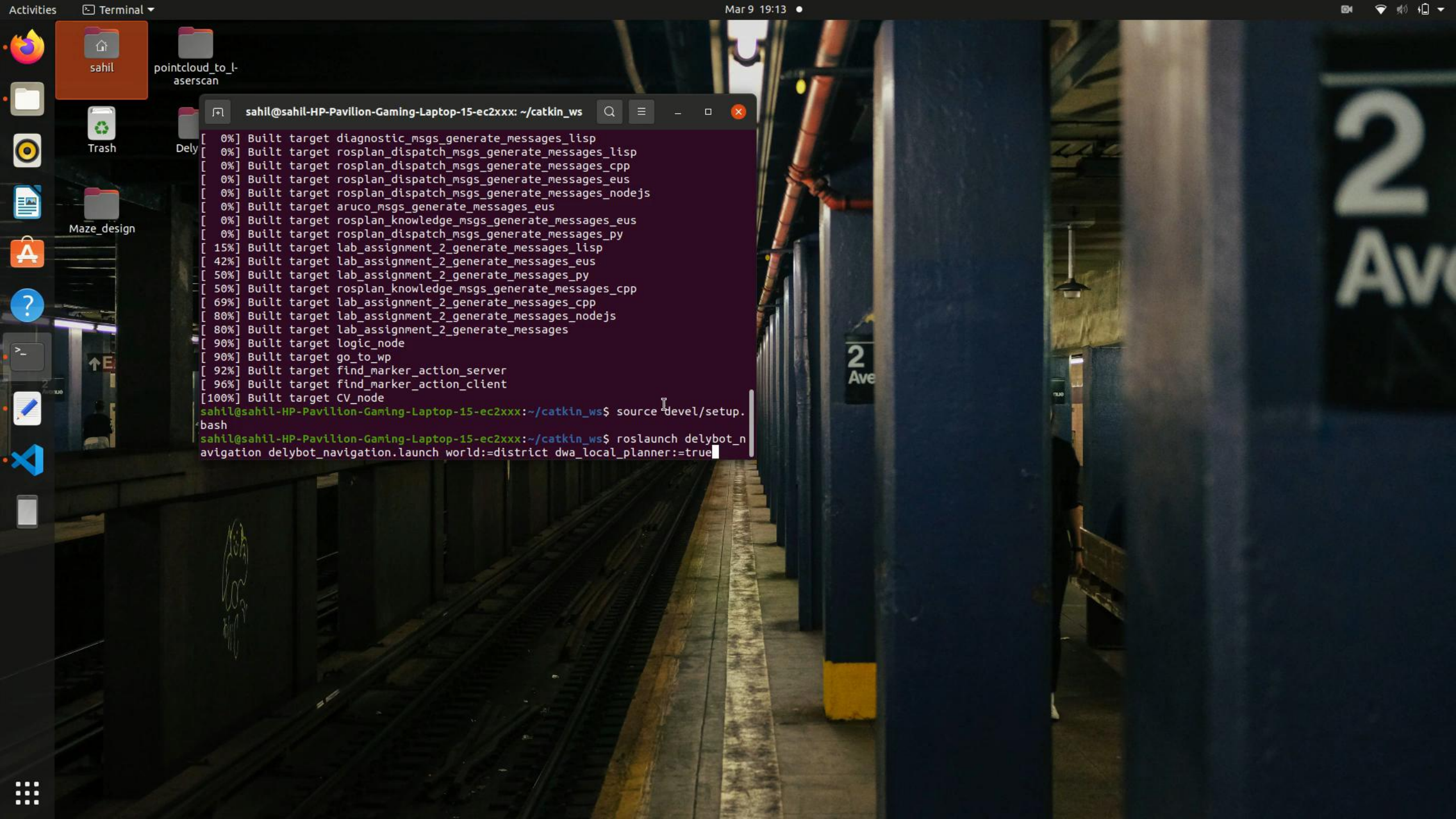
- This is the static programming of the robot where the robot movement is defined by giving waypoints at the start.
- The major obstacle detection is done by 3D point cloud sensor and the camera is there for region of interest.
- The two methods described are
 - Move Base package using DWA and trajectory planning
 - The base programming using A* along with local and global map
- The simulations are done in the gazebo and RVIZ for initial testing. The sensor data collection along with final testing was done in the UGV with some major changes.
- Along with that airport tug designing and PLANSYS2 in ROS 2.

ROS INSTALLATION AND CAMERA INTEGRATION TO TUG IN RVIZ+GAZEBO

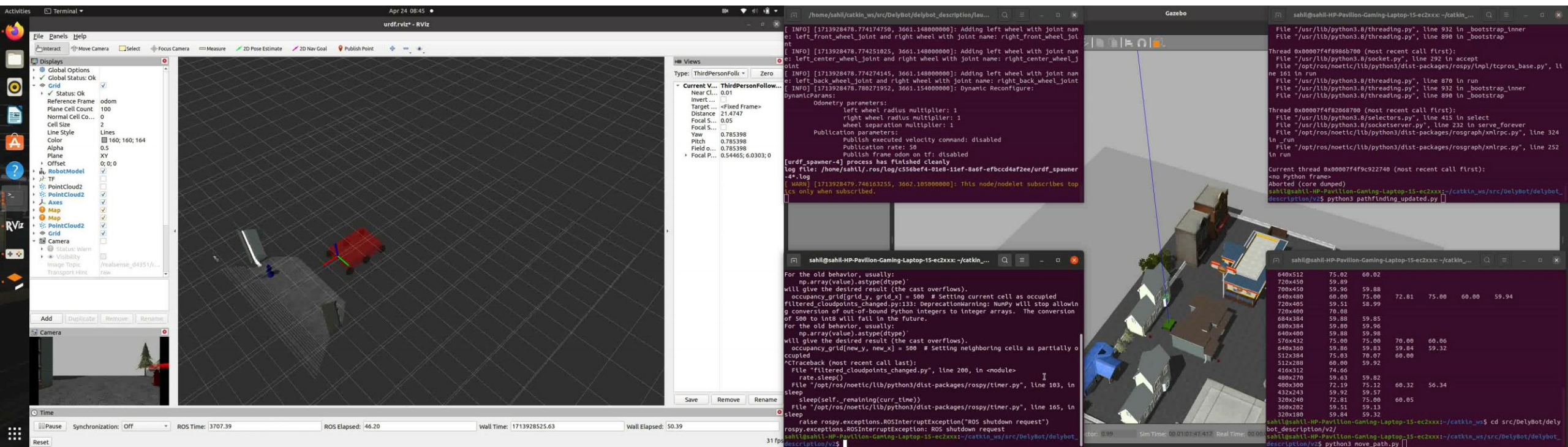


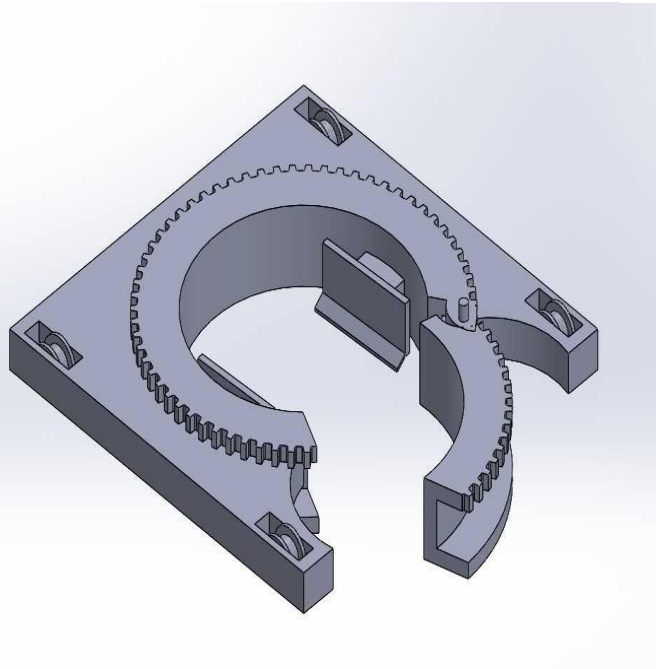
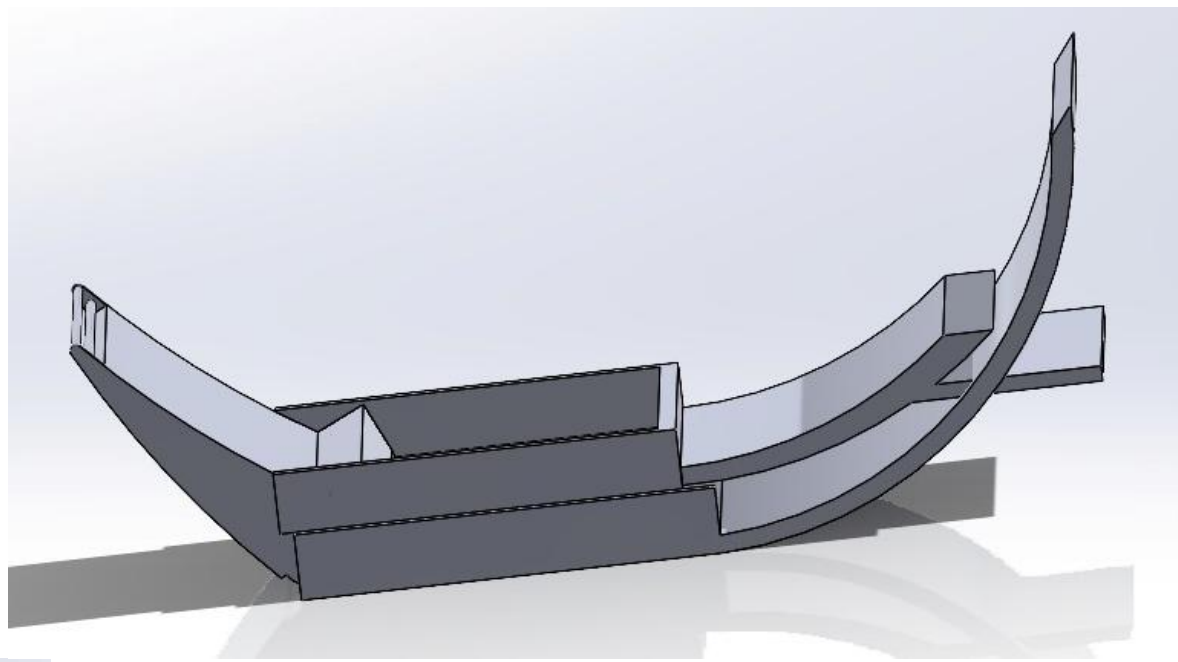
Path planning using A* and SLAM



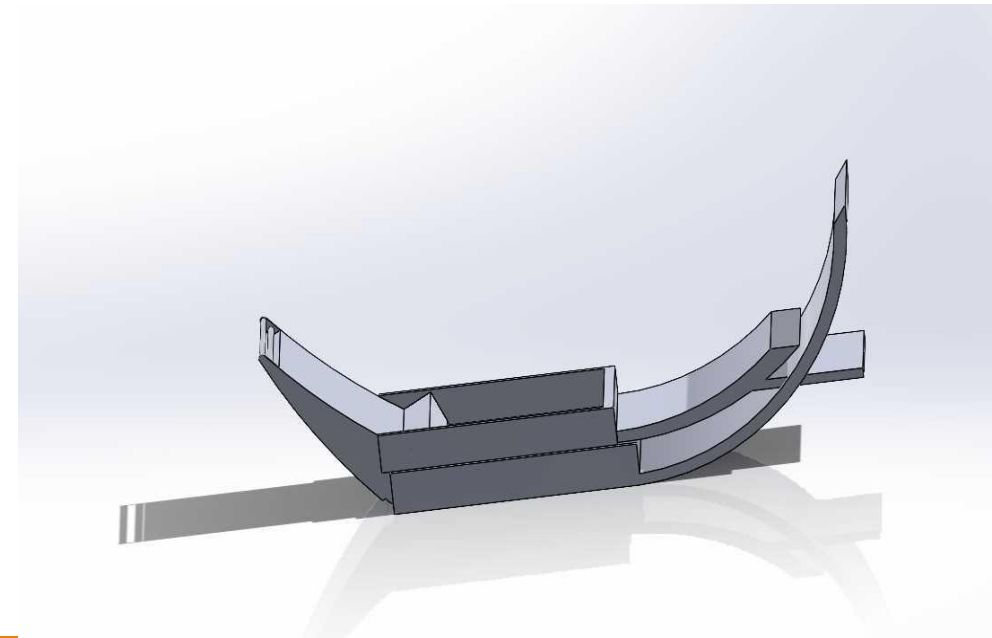


Path-planning algorithm



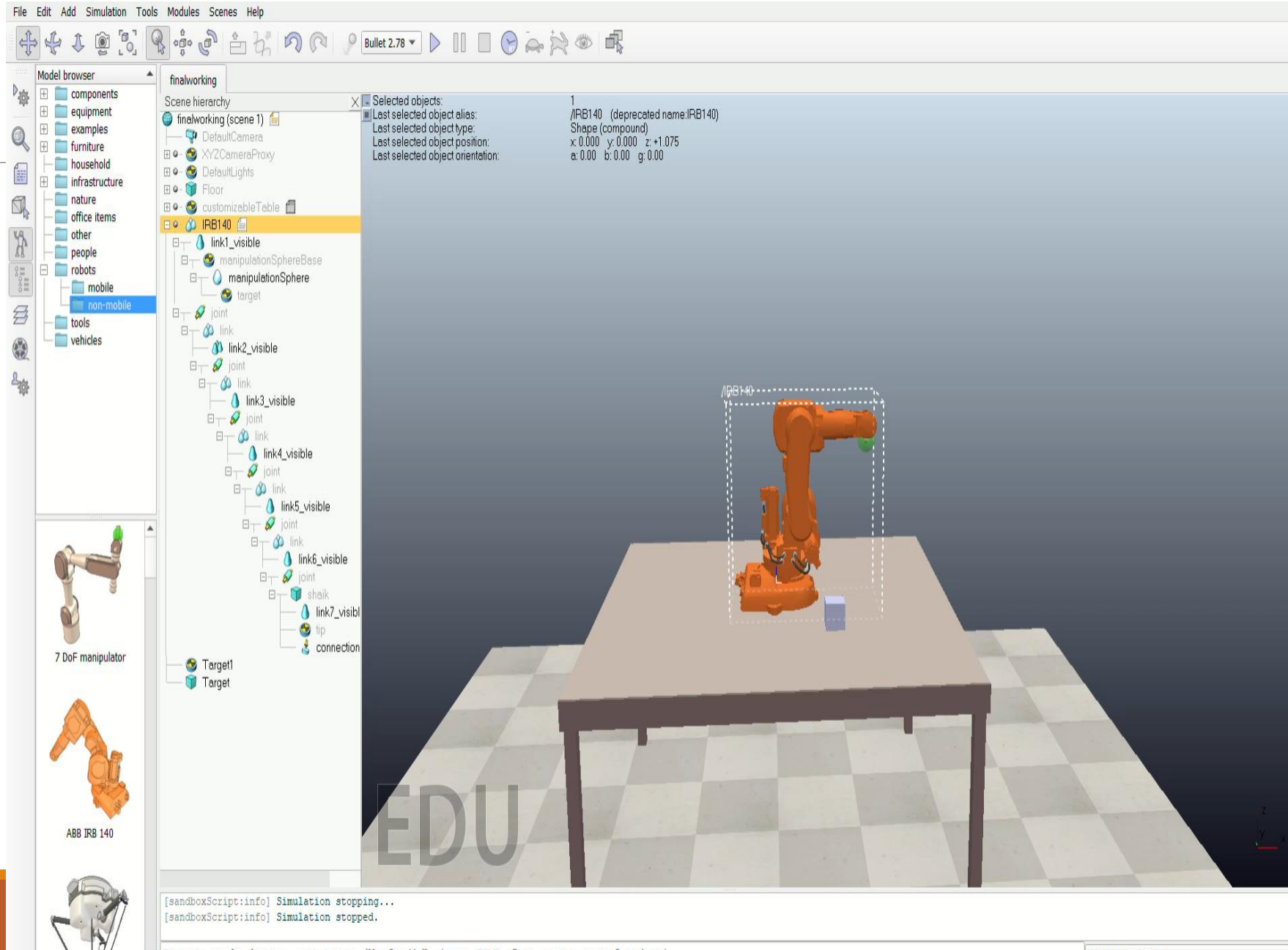


AIRPORT TUG
DESIGNS

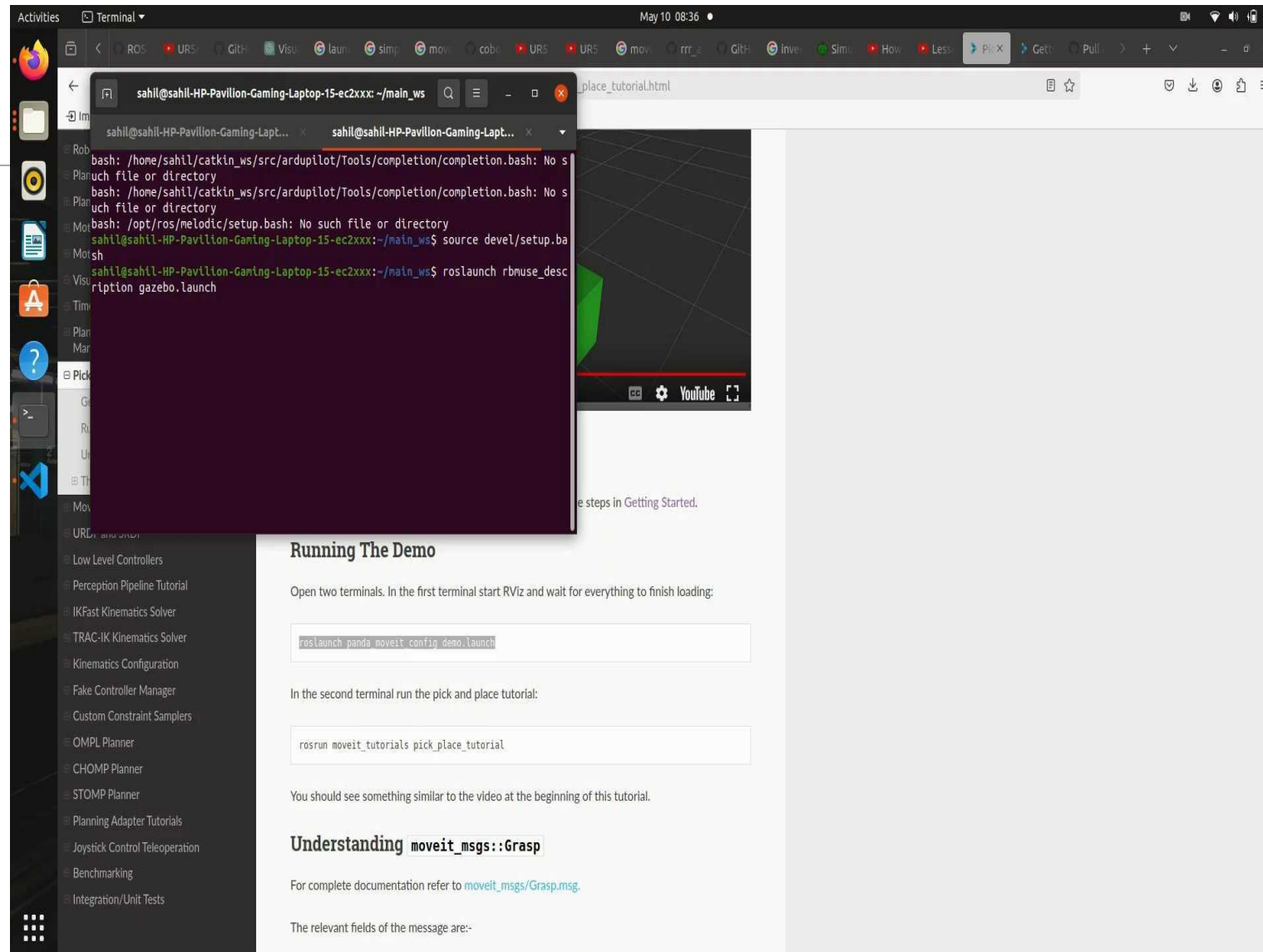


Other projects

Robotics arm forward kinematics Vrep



Robotics arm Moveit simulation



The screenshot shows a Linux desktop environment. A terminal window is open, displaying the following commands and output:

```
sahil@sahil-HP-Pavilion-Gaming-Laptop-15-ec2xxx: ~/main_ws
bash: /home/sahil/catkin_ws/src/ardupilot/Tools/completion/completion.bash: No such file or directory
bash: /home/sahil/catkin_ws/src/ardupilot/Tools/completion/completion.bash: No such file or directory
bash: /opt/ros/melodic/setup.bash: No such file or directory
sahil@sahil-HP-Pavilion-Gaming-Laptop-15-ec2xxx:~/main_ws$ source devel/setup.bash
sahil@sahil-HP-Pavilion-Gaming-Laptop-15-ec2xxx:~/main_ws$ roslaunch rbmuse_description gazebo.launch
```

A web browser window is open, displaying a tutorial page titled "pick_place_tutorial.html". The page content includes:

Running The Demo

Open two terminals. In the first terminal start RViz and wait for everything to finish loading:

```
roslaunch panda_moveit_config demo.launch
```

In the second terminal run the pick and place tutorial:

```
roslaunch moveit_tutorials pick_place_tutorial
```

You should see something similar to the video at the beginning of this tutorial.

Understanding `moveit_msgs::Grasp`

For complete documentation refer to [moveit_msgs/Grasp.msg](#).

The relevant fields of the message are:-

Reinforcement learning

colab.research.google.com/drive/1Q_1msUh9puaRuTfmYnXfg6d5q...

CliffWalking_game.ipynb

File Edit View Insert Runtime Tools Help All changes saved

+ Code + Text

RAM
Disk

Gemini

```
import numpy as np (module) pyplot
import matplotlib.pyplot as plt
from IPython import display as ipythondisplay
import time
```

```
[4] from gym import envs
    from gym.spaces import Discrete, Box
    all_envs = envs.registry.all()
    env_ids = [env_spec.id for env_spec in all_envs]
    print(sorted(env_ids))
```

```
['Acrobot-v1', 'Ant-v2', 'Ant-v3', 'Ant-v4', 'BipedalWalker-v3', 'BipedalWalkerHardcore-v3', 'Blackjack-v1', 'CarRacing-v2', 'CartPole-v0', 'CartPole-v1', 'CliffWalking-v0', 'FrozenLake-v0', 'GymEnv-v0', 'HalfCheetah-v2', 'Hopper-v2', 'Humanoid-v2', 'Humanoid-v3', 'InvertedPendulum-v2', 'InvertedPendulum-v3', 'LunarLander-v2', 'LunarLander-v3', 'MountainCar-v0', 'MountainCar-v2', 'Pendulum-v1', 'Reacher-v2', 'Seaquest-v0', 'Seaquest-v1', 'TigerLand-v3', 'TigerLand-v4', 'TigerLand-v5', 'TigerLand-v6', 'TigerLand-v7', 'TigerLand-v8', 'TigerLand-v9', 'TigerLand-v10', 'TigerLand-v11', 'TigerLand-v12', 'TigerLand-v13', 'TigerLand-v14', 'TigerLand-v15', 'TigerLand-v16', 'TigerLand-v17', 'TigerLand-v18', 'TigerLand-v19', 'TigerLand-v20', 'TigerLand-v21', 'TigerLand-v22', 'TigerLand-v23', 'TigerLand-v24', 'TigerLand-v25', 'TigerLand-v26', 'TigerLand-v27', 'TigerLand-v28', 'TigerLand-v29', 'TigerLand-v30', 'TigerLand-v31', 'TigerLand-v32', 'TigerLand-v33', 'TigerLand-v34', 'TigerLand-v35', 'TigerLand-v36', 'TigerLand-v37', 'TigerLand-v38', 'TigerLand-v39', 'TigerLand-v40', 'TigerLand-v41', 'TigerLand-v42', 'TigerLand-v43', 'TigerLand-v44', 'TigerLand-v45', 'TigerLand-v46', 'TigerLand-v47', 'TigerLand-v48', 'TigerLand-v49', 'TigerLand-v50', 'TigerLand-v51', 'TigerLand-v52', 'TigerLand-v53', 'TigerLand-v54', 'TigerLand-v55', 'TigerLand-v56', 'TigerLand-v57', 'TigerLand-v58', 'TigerLand-v59', 'TigerLand-v60', 'TigerLand-v61', 'TigerLand-v62', 'TigerLand-v63', 'TigerLand-v64', 'TigerLand-v65', 'TigerLand-v66', 'TigerLand-v67', 'TigerLand-v68', 'TigerLand-v69', 'TigerLand-v70', 'TigerLand-v71', 'TigerLand-v72', 'TigerLand-v73', 'TigerLand-v74', 'TigerLand-v75', 'TigerLand-v76', 'TigerLand-v77', 'TigerLand-v78', 'TigerLand-v79', 'TigerLand-v80', 'TigerLand-v81', 'TigerLand-v82', 'TigerLand-v83', 'TigerLand-v84', 'TigerLand-v85', 'TigerLand-v86', 'TigerLand-v87', 'TigerLand-v88', 'TigerLand-v89', 'TigerLand-v90', 'TigerLand-v91', 'TigerLand-v92', 'TigerLand-v93', 'TigerLand-v94', 'TigerLand-v95', 'TigerLand-v96', 'TigerLand-v97', 'TigerLand-v98', 'TigerLand-v99']
/usr/local/lib/python3.10/dist-packages/gym/envs/registration.py:421: UserWarning: WARN: The `registry.all` method is deprecated. Please use `registry.values` instead.
  logger.warn(
```

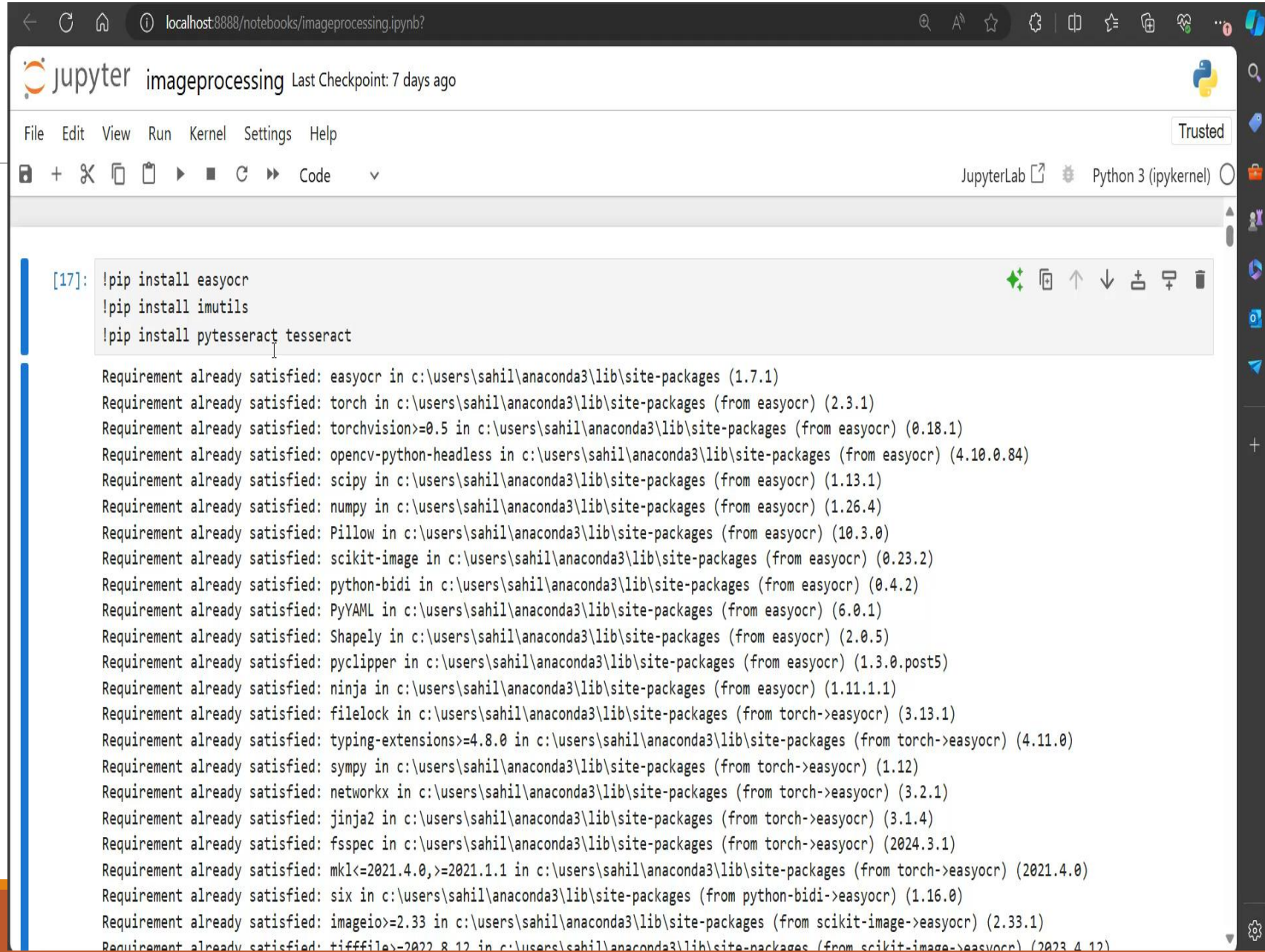
```
[5] import gym
    env = gym.make("CliffWalking-v0")
```

```
/usr/local/lib/python3.10/dist-packages/gym/core.py:317: DeprecationWarning: WARN: Initializing wrapper in old step API which returns one bool instead of two. It is recommended to set deprecation(
  deprecation(
/usr/local/lib/python3.10/dist-packages/gym/wrappers/step_api_compatibility.py:39: DeprecationWarning: WARN: Initializing environment in old step API which returns one bool instead of deprecation(
  deprecation(
```

```
[6] class QlearningAgent:
    def __init__(self, n_states, n_act, e_grade=0.1, gamma=0.9, lr=0.1):
        self.Q = np.zeros((n_states, n_act))
        self.e_grade = e_grade
        self.gamma = gamma
```

completed at 12:36 PM

Image processing and Identification



The screenshot shows a JupyterLab interface in a web browser. The address bar indicates the URL is `localhost:8888/notebooks/imageprocessing.ipynb?`. The JupyterLab header shows the name of the notebook as `imageprocessing` and the last checkpoint as `7 days ago`. The interface includes a menu bar with `File`, `Edit`, `View`, `Run`, `Kernel`, `Settings`, and `Help`. Below the menu bar is a toolbar with icons for file operations and a `Code` dropdown menu. The right side of the interface shows the `Trusted` status and the `Python 3 (ipykernel)` environment. The main area displays a terminal window with the following content:

```
[17]: !pip install easyocr
      !pip install imutils
      !pip install pytesseract tesseract

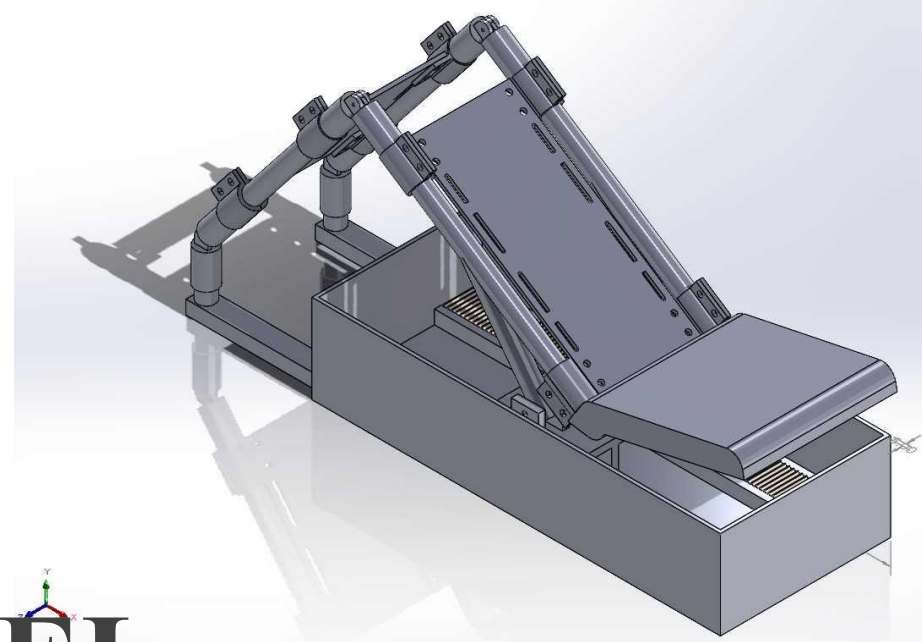
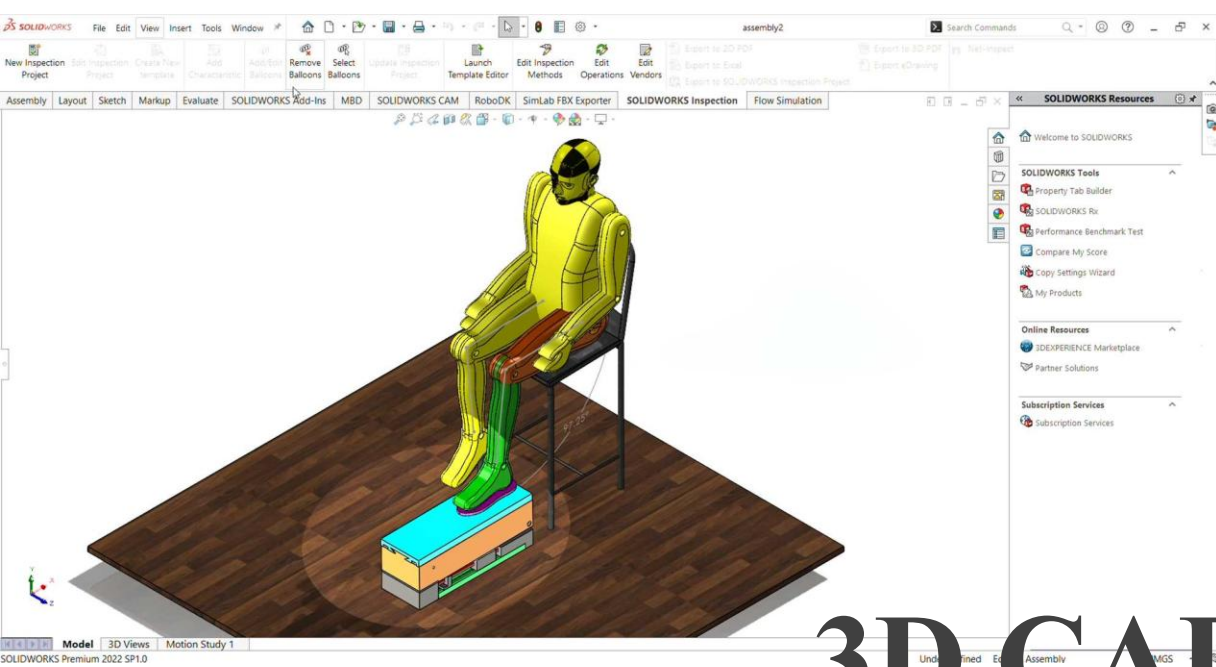
Requirement already satisfied: easyocr in c:\users\sahil\anaconda3\lib\site-packages (1.7.1)
Requirement already satisfied: torch in c:\users\sahil\anaconda3\lib\site-packages (from easyocr) (2.3.1)
Requirement already satisfied: torchvision>=0.5 in c:\users\sahil\anaconda3\lib\site-packages (from easyocr) (0.18.1)
Requirement already satisfied: opencv-python-headless in c:\users\sahil\anaconda3\lib\site-packages (from easyocr) (4.10.0.84)
Requirement already satisfied: scipy in c:\users\sahil\anaconda3\lib\site-packages (from easyocr) (1.13.1)
Requirement already satisfied: numpy in c:\users\sahil\anaconda3\lib\site-packages (from easyocr) (1.26.4)
Requirement already satisfied: Pillow in c:\users\sahil\anaconda3\lib\site-packages (from easyocr) (10.3.0)
Requirement already satisfied: scikit-image in c:\users\sahil\anaconda3\lib\site-packages (from easyocr) (0.23.2)
Requirement already satisfied: python-bidi in c:\users\sahil\anaconda3\lib\site-packages (from easyocr) (0.4.2)
Requirement already satisfied: PyYAML in c:\users\sahil\anaconda3\lib\site-packages (from easyocr) (6.0.1)
Requirement already satisfied: Shapely in c:\users\sahil\anaconda3\lib\site-packages (from easyocr) (2.0.5)
Requirement already satisfied: pyclicker in c:\users\sahil\anaconda3\lib\site-packages (from easyocr) (1.3.0.post5)
Requirement already satisfied: ninja in c:\users\sahil\anaconda3\lib\site-packages (from easyocr) (1.11.1.1)
Requirement already satisfied: filelock in c:\users\sahil\anaconda3\lib\site-packages (from torch->easyocr) (3.13.1)
Requirement already satisfied: typing-extensions>=4.8.0 in c:\users\sahil\anaconda3\lib\site-packages (from torch->easyocr) (4.11.0)
Requirement already satisfied: sympy in c:\users\sahil\anaconda3\lib\site-packages (from torch->easyocr) (1.12)
Requirement already satisfied: networkx in c:\users\sahil\anaconda3\lib\site-packages (from torch->easyocr) (3.2.1)
Requirement already satisfied: Jinja2 in c:\users\sahil\anaconda3\lib\site-packages (from torch->easyocr) (3.1.4)
Requirement already satisfied: fsspec in c:\users\sahil\anaconda3\lib\site-packages (from torch->easyocr) (2024.3.1)
Requirement already satisfied: mkl<=2021.4.0,>=2021.1.1 in c:\users\sahil\anaconda3\lib\site-packages (from torch->easyocr) (2021.4.0)
Requirement already satisfied: six in c:\users\sahil\anaconda3\lib\site-packages (from python-bidi->easyocr) (1.16.0)
Requirement already satisfied: imageio>=2.33 in c:\users\sahil\anaconda3\lib\site-packages (from scikit-image->easyocr) (2.33.1)
Requirement already satisfied: tifffile>=2022.8.12 in c:\users\sahil\anaconda3\lib\site-packages (from scikit-image->easyocr) (2023.4.12)
```



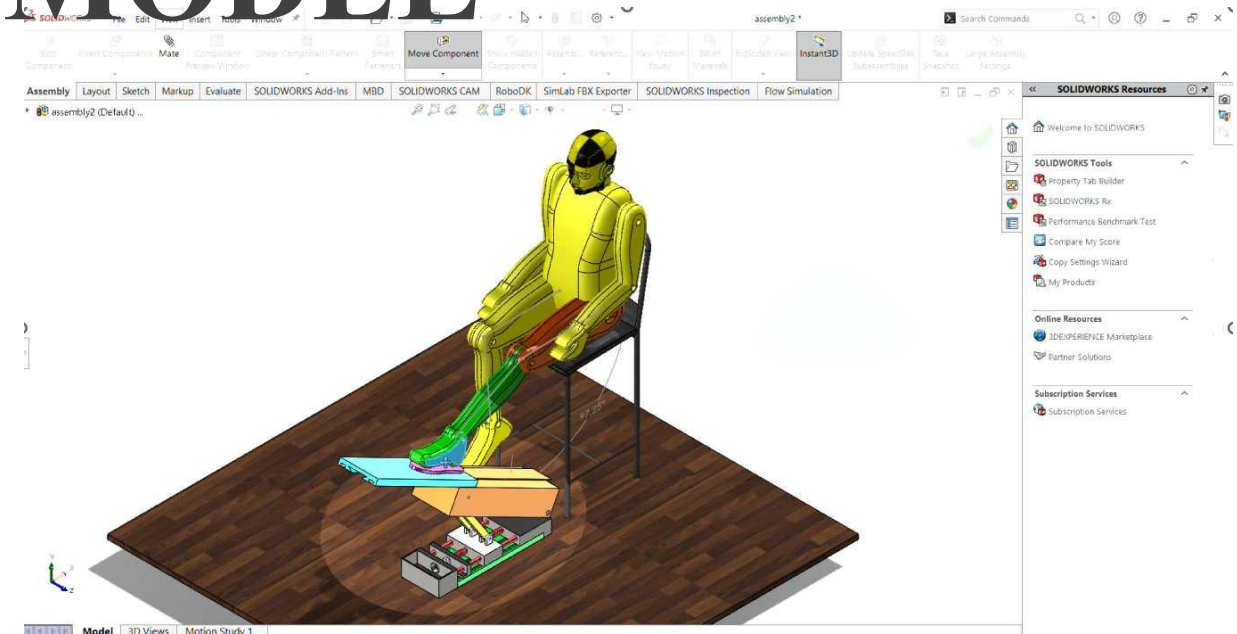
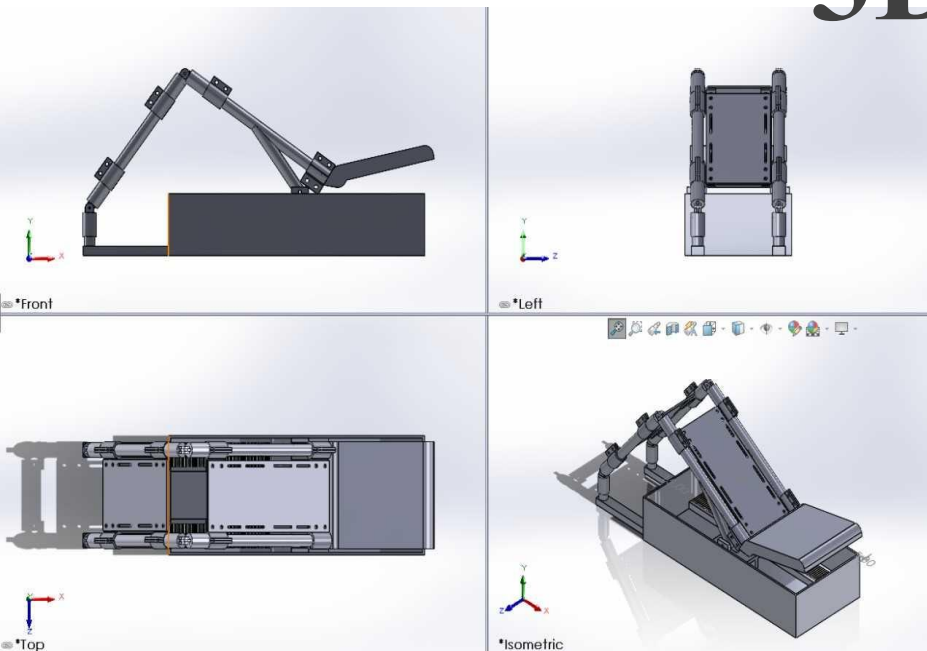

Design and development of knee rehabilitation robot

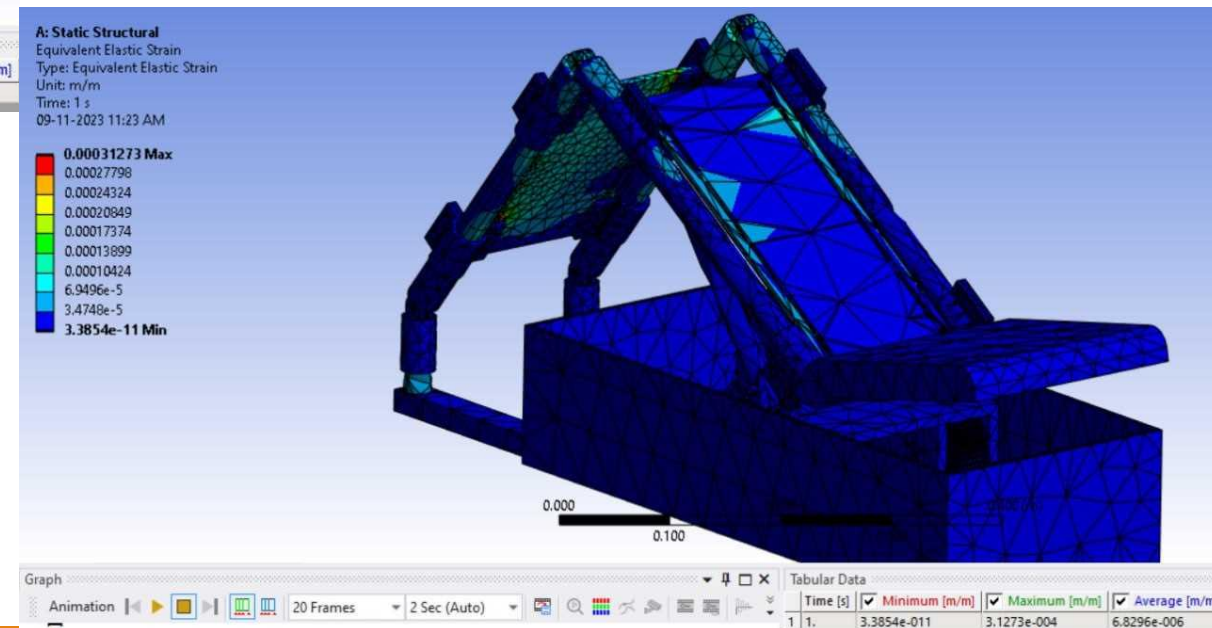
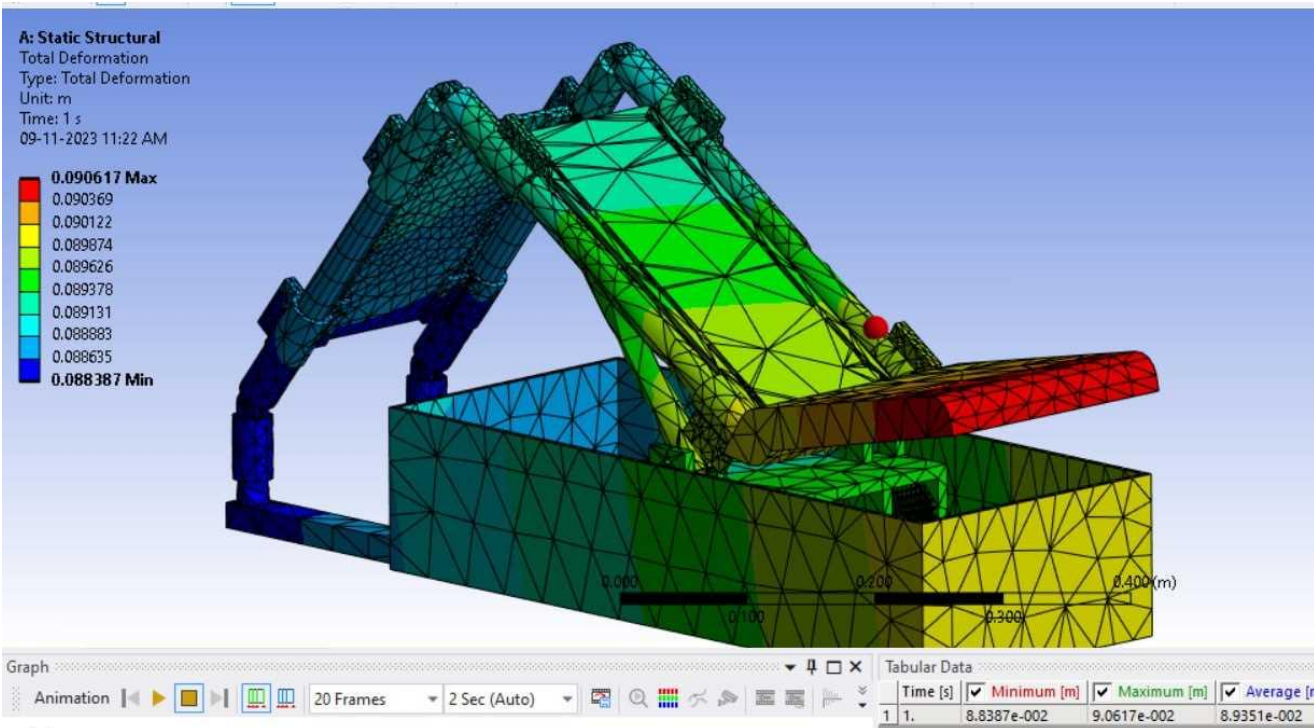
Design and development of knee rehabilitation robot

- The robot is designed to give physiotherapy in the absence of the doctors. The robot's mechanical design is to hold 2 degrees of motion and give an arc movement instead of an X-Y movement.
- The robot had to undergo extensive research for available designs and their advantages & disadvantages. To find the best optimal design to mimic human action.
- AI algorithm was applied to check the patients had knee osteosis, using the convolution algorithm.
- One design is going through a patent and the other is going to be published and hopefully be patent in the coming future



3D CAD MODEL



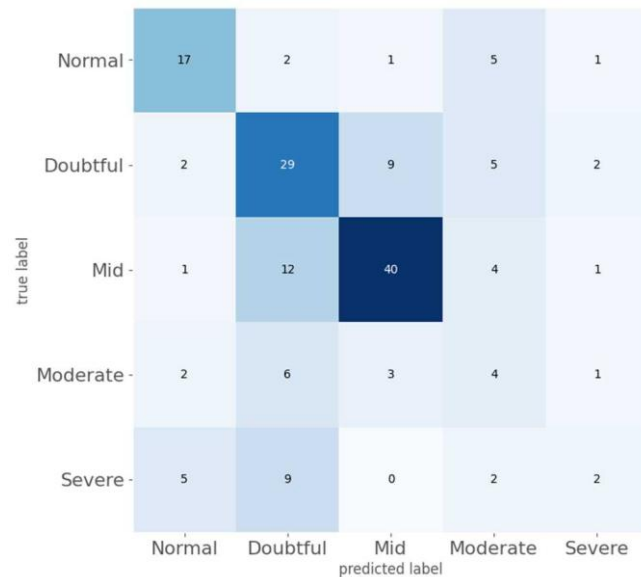


AI ALGORITHM OUTPUT(Relu activation)



test loss: 3.478864908218384 %

test accuracy: 0.5575757622718811 %



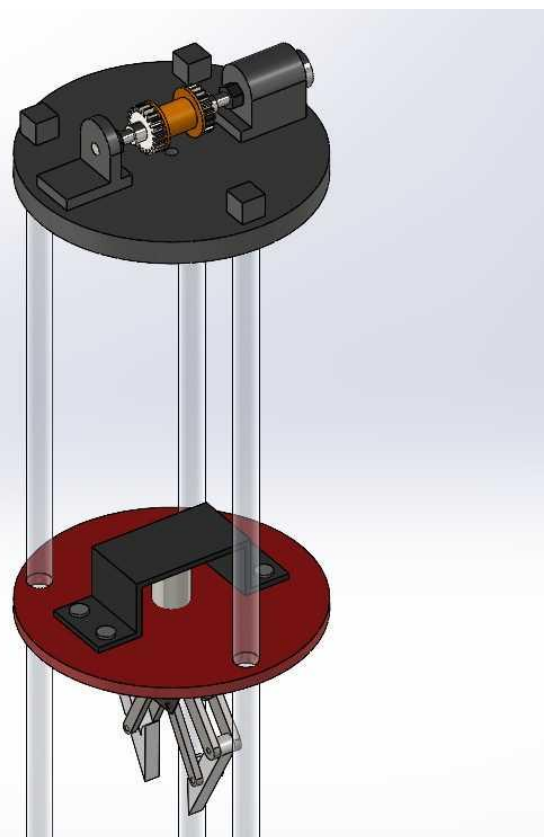
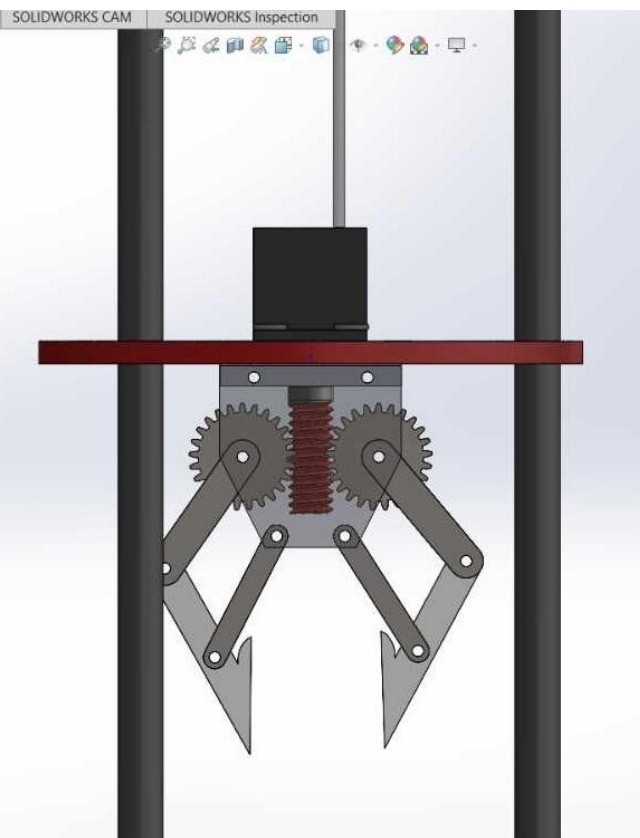
Activation Function	Epoch	Test loss	Test accuracy
Relu	100	3.48	0.56
Relu	50	2.82	0.53
Linear	100	3.71	0.50
Linear	50	3.11	0.50
LeakyRelu	100	4.58	0.52
LeakyRelu	50	3.08	0.52

Design and Development of Borewell Rescue System

Design and Development of Borewell Rescue System

- The borewell system was designed to save children who fall into the borewell. The robot has soft grippers to rescue them without harming them
- The design was done keeping the dimensions accuracy and the joint functionality.
- The designing major motto was to keep the design simple and affordable.
- The major designs were taken to be as market-available and programable content was also made to design and upload within minutes so as of not to waste time.
- The final and initial designs as in the below slide along with simulation.

BOREWELL RESCUE ROBOT



Initial design



Design and development of a homogeneous swarm of cooperative robots in a virtual environment

Design and development of a homogeneous swarm of cooperative robots in a virtual environment

- The homogeneous robot communications were explored and studied in detail the algorithms and formulas related to the swarm robotics.
- The Webot software provides a similar experience as of ROS along with prebuilt packages to learn to code in whichever language is comfortable.
- The robots are communicating with each other using the world node as the medium the speed is getting communicating with each other hence movement is possible
- The robot adding the cumulative movement in the X-Y plane is done and the output was not satisfactory as it needed interrupt control to avoid obstacles first.



- > WorldInfo
- > Viewpoint
- > TexturedBackground
- > TexturedBackgroundLight
- > RectangleArena "rectangle arena"
- > DEF master TurtleBot3Burger
- > DEF slave TurtleBot3Burger

Selection: RectangleArena (Solid)

Node Position Velocity

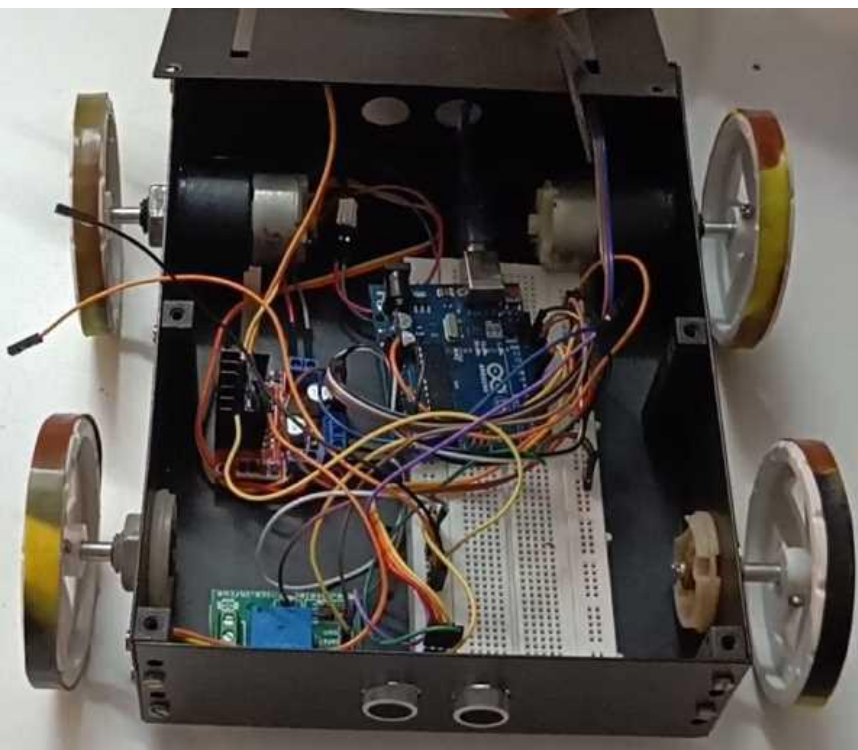
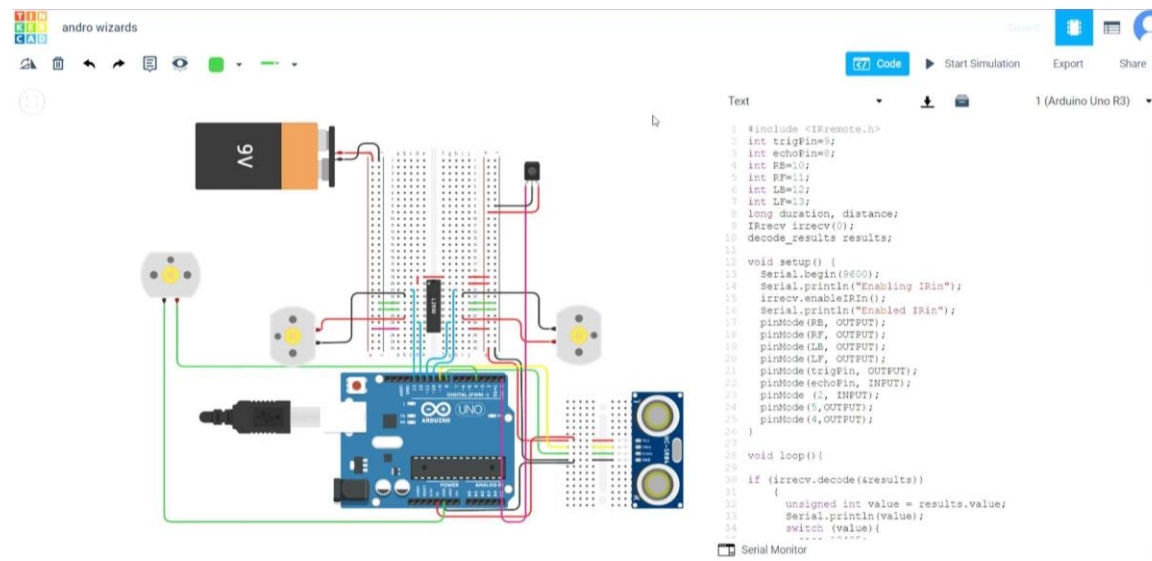
DEF: 

```
my_controller3.py x my_controller1.py x
1 from controller import Robot, Motor
2 TIME_STEP = 64
3 MAX_SPEED = 6.28
4 robot = Robot()
5 leftMotor = robot.getDevice('left wheel motor')
6 rightMotor = robot.getDevice('right wheel motor')
7 leftMotor.setPosition(float('inf'))
8 rightMotor.setPosition(float('inf'))
9 leftMotor.setVelocity(0.1 * MAX_SPEED)
10 rightMotor.setVelocity(0.1 * MAX_SPEED)
11
12 while robot.step(TIME_STEP) != -1:
13     pass
```

Designing and developing of autonomous sanitizing robot

Designing and developing of autonomous sanitizing robot

- The robot was visioned during the lockdown time to give to the shopkeepers to sanitize their shops at regular intervals to ensure sanitized shops.
- The robot uses a basic avoidance program by using ultrasonic sensor data.
- The robot has an IR sensor which is limited by distance to detect objects for the spray to initialize and spray on the cupboard of the shops.
- The robot chassis was tested to hold the water tank and the program simulation was made in Tinkercad for simulation and code testing.



Sanitization robot



THANK YOU



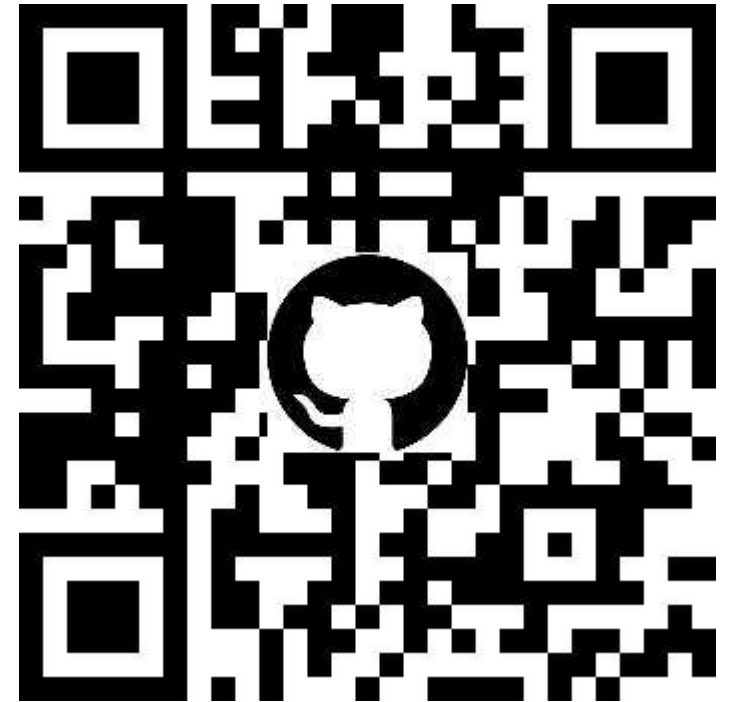
www.linkedin.com/in/sahil-shaik-8bb309219



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Phone: +91 7337013571



<https://github.com/sahil86469>